Bass Real Analysis Solutions

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This website contains my solutions to selected exercises from Richard Bass's *Real Analysis for Graduate Students v5.0*.

2.1 About this project

The goal of this project is to provide clear and rigorous solutions to graduate-level real analysis exercises.

2.2 Author

Jae-Yun Lee

- B.S. in Mathematical Sciences, KAIST
- M.S. and Ph.D. Track in Economics, KAIST

For more information, see the About page.

3 About

4 Jae-Yun Lee

I am an independent researcher and musician from South Korea.

4.1 Education

- B.S. in Mathematical Sciences, KAIST
- M.S. and Ph.D. Track in Economics, Department of Management Engineering, KAIST

4.2 Research

Selected publication:

> Jae-Yun Lee, Jiwoong Shin, and Jungju Yu
>
> "Communicating Attribute Importance under Competition"
>
> *Marketing Science*, 45(3), 534–555, 2026.

4.3 Music

Outside academia, I work as an independent musician under the name **Jayhat**.

My work combines hip-hop, R&B, and classical music, with a particular interest in orchestral composition and piano performance.

- Instagram: (**jayhat3210?**)

Part I

Chapter 2

5 Exercise 2.1

5.1 Solution

Consider $X = \{1, 2, 3\}$ and $\mathcal{M} = \{\emptyset, \{1\}, \{2\}, \{3\}, X\}$.

It is straightforward to verify that $\mathcal{M}$ is a monotone class. However, $\mathcal{M}$ is not a σ-algebra, since

$$\{1\} \cup \{2\} = \{1, 2\} \notin \mathcal{M}.$$

6 Exercise 2.2

6.1 Solution

Consider $X = \{1, 2, 3, 4\}$, and

$$\mathcal{A}_1 = \{\emptyset, \{1\}, \{2, 3, 4\}, X\}, \quad \mathcal{A}_2 = \{\emptyset, \{1, 2, 3\}, \{4\}, X\}.$$

It is straightforward to verify that $\mathcal{A}_1$ and $\mathcal{A}_2$ are σ-algebras. However, $\mathcal{A}_1 \cup \mathcal{A}_2$ is not a σ-algebra, since

$$\{1\} \cup \{4\} = \{1, 4\} \notin \mathcal{A}_1 \cup \mathcal{A}_2.$$

7 Exercise 2.3

7.1 Solution

Consider $X = \mathbb{N}$ and $\mathcal{A}_n = \sigma\Big(\{\{1\}, \{2\}, \cdots, \{n\}\}\Big)$.

Note that $\mathcal{A}_n$ is a σ-algebra by construction. Moreover, since

$$\{\{1\}, \{2\}, \cdots, \{n\}\} \subset \{\{1\}, \{2\}, \cdots, \{n+1\}\},$$

we have $\mathcal{A}_n \subset \mathcal{A}_{n+1}$.

First, we establish a useful property of $\mathcal{A}_n$.

For each $n \in \mathbb{N}$, define

$$T_n = \{n+1, n+2, \cdots\} \quad \text{and} \quad \mathcal{B}_n = \{E \subset \mathbb{N} : T_n \subset E \ \text{ or } \ E \cap T_n = \emptyset\}.$$

We claim that $\mathcal{B}_n$ is a σ-algebra.

(1) First, $\emptyset \in \mathcal{B}_n$, since $\emptyset \cap T_n = \emptyset$.

(2) Second, suppose that $E \in \mathcal{B}_n$.

If $T_n \subset E$, then $E^c \cap T_n = \emptyset$. And if $E \cap T_n = \emptyset$, then $T_n \subset E^c$.

Thus, we have $E^c \in \mathcal{B}_n$.

(3) Third, suppose that $E_k \in \mathcal{B}_n$ for $k = 1, 2, \cdots$.

If there exists k such that $T_n \subset E_k$, then $T_n \subset \bigcup_k E_k$. Thus, we have $\bigcup_k E_k \in \mathcal{B}_n$.

Otherwise, $E_k \cap T_n = \emptyset$ for all k. Hence,

$$\Big(\bigcup_k E_k\Big) \cap T_n = \bigcup_k (E_k \cap T_n) = \emptyset.$$

Therefore, $\mathcal{B}_n$ is a σ-algebra.

Next, since $\{1\}, \dots, \{n\} \in \mathcal{B}_n$, we conclude that

$$\mathcal{A}_n = \sigma\Big(\{\{1\}, \{2\}, \cdots, \{n\}\}\Big) \subset \mathcal{B}_n.$$

Consequently, for every $E \in \mathcal{A}_n$, we have $E \cap T_n = \emptyset$ or $T_n \subset E$.

Using this property, we will show that $\bigcup_k \mathcal{A}_k$ is not a σ-algebra.

Consider the set

$$\{2, 4, 6, \cdots\} = \bigcup_n \{2n\}.$$

For any $n \in \mathbb{N}$, we have

$$\{2, 4, 6, \cdots\} \cap T_n \neq \emptyset \quad \text{and} \quad T_n \not\subset \{2, 4, 6, \cdots\}.$$

Hence, $\{2, 4, 6, \cdots\} \notin \mathcal{A}_n$ for any $n \in \mathbb{N}$, which implies that $\{2, 4, 6, \cdots\} \notin \bigcup_k \mathcal{A}_k$.

On the other hand, $\{2n\} \in \mathcal{A}_{2n} \subset \bigcup_k \mathcal{A}_k$ for each $n \in \mathbb{N}$.

Therefore, $\bigcup_k \mathcal{A}_k$ is not closed under countable unions, and therefore is not a σ-algebra.

8 Exercise 2.4

8.1 Solution

(I think it is $A_j \in \mathcal{M}_j$ in exercise, not $A_j \in \mathcal{M}$)

Consider $A_n = \{2, 4, \cdots, 2n\}$ and $\mathcal{M}_n = \{\emptyset, A_n, \mathbb{N}\}$. It is straightforward to verify that

(1) $\mathcal{M}_n$ is a monotone class, and

(2) $A_n \uparrow A = \{2, 4, 6, \cdots\}$.

However, $A \neq A_n$ for every n, so, $A \notin \mathcal{M}$.

9 Exercise 2.5

9.1 Solution

We show that $\mathcal{B}$ is a σ-algebra.

(1) $\emptyset \in \mathcal{B}$

Since $\emptyset \in \mathcal{A}$, we have $f^{-1}(\emptyset) \in \mathcal{B}$. And we can say that $f^{-1}(\emptyset) = \emptyset$.

To prove this, suppose not. Then there exists $x \in X$, such that $x \in f^{-1}(\emptyset)$. That is, $f(x) \in \emptyset$. But it is a contradiction.

(2) $X \in \mathcal{B}$

Since $Y \in \mathcal{A}$, we have $f^{-1}(Y) \in \mathcal{B}$. And we can say that $f^{-1}(Y) = X$.

To prove this, suppose not. Then there exists $x \in X$, such that $x \notin f^{-1}(Y)$. That is, $f(x) \notin Y$. But it is a contradiction.

(3) If $B \in \mathcal{B}$, then $B^c \in \mathcal{B}$

Consider $B \in \mathcal{B}$. Then there exists $A \in \mathcal{A}$, such that $f^{-1}(A) = B$. Since

$$f^{-1}(A^c) = X \setminus f^{-1}(A) = B^c,$$

and $A^c \in \mathcal{A}$, we can conclude that $B^c \in \mathcal{B}$.

(4) If $B_1, B_2, ... \in \mathcal{B}$, then $\bigcup_i B_i \in \mathcal{B}$

Consider $B_1, B_2, ... \in \mathcal{B}$ and corresponding $A_1, A_2, ... \in \mathcal{A}$. Since

$$\bigcup_i B_i = \bigcup_i f^{-1}(A_i) = f^{-1}\Big(\bigcup_i A_i\Big),$$

and $\bigcup_i A_i \in \mathcal{A}$, we can conclude that $\bigcup_i B_i \in \mathcal{B}$.

10 Exercise 2.6

10.1 Solution

Suppose, for contradiction, that $\mathcal{A}$ is finite.

Since $X \neq \emptyset$ and $X \in \mathcal{A}$, the assumed property implies that X can be written as a disjoint union of two nonempty sets in $\mathcal{A}$.

Choose one of them and call it A_1. Now apply the same property to A_1. Then A_1 can be written as a disjoint union of two nonempty sets in $\mathcal{A}$.

Choose one of them and call it A_2. Since the other piece is also nonempty, we have

$$A_2 \subsetneq A_1.$$

Repeating this argument, we obtain a strictly decreasing sequence

$$X \supsetneq A_1 \supsetneq A_2 \supsetneq A_3 \supsetneq \cdots$$

of sets in $\mathcal{A}$.

This is impossible if $\mathcal{A}$ is finite. Hence $\mathcal{A}$ is infinite.

By Exercise 2.8, every infinite σ-algebra is uncountable. Therefore, $\mathcal{A}$ is uncountable.

11 Exercise 2.7

11.1 Solution

We show that $\mathcal{A}$ is a σ-algebra.

(1) $\emptyset \in \mathcal{A}$

Since $\chi_\emptyset(x) = 0$ for every $x \in X$, it is a constant function. Hence,

$$\chi_\emptyset \in \mathcal{F}.$$

Therefore, $\emptyset \in \mathcal{A}$.

(2) $X \in \mathcal{A}$

Since $\chi_X(x) = 1$ for evert $x \in X$, it is a constant function. Hence,

$$\chi_X \in \mathcal{F}.$$

Therefore, $X \in \mathcal{A}$.

(3) If $A \in \mathcal{A}$, then $A^c \in \mathcal{A}$

Let $A \in \mathcal{A}$. Then $\chi_A \in \mathcal{F}$. Since

$$\chi_{A^c} = 1 - \chi_A,$$

and since $1 \in \mathcal{F}$, it follows that

$$\chi_{A^c} \in \mathcal{F}.$$

Therefore, $A^c \in \mathcal{A}$.

(4) If $A_1, A_2, ... \in \mathcal{A}$, then $\bigcup_i A_i \in \mathcal{A}$

First, if $\chi_A, \chi_B \in \mathcal{F}$, then

$$\chi_{A\cup B} = \chi_A + \chi_B - \chi_A\chi_B.$$

Since $\chi_A, \chi_B \in \mathcal{F}$, we have

$$\chi_{A\cup B} \in \mathcal{F}.$$

Thus, for every $n \in \mathbb{N}$,
$$f_n := \chi_{\bigcup_{i=1}^n A_i} \in \mathcal{F}.$$

Moreover,
$$f_n(x) \to \chi_{\bigcup_i A_i}(x) \quad \text{for every } x \in X.$$

Since $\mathcal{F}$ is closed under pointwise limits, we have
$$\chi_{\bigcup_i A_i} \in \mathcal{F}.$$

Therefore,
$$\bigcup_i A_i \in \mathcal{A}.$$

Hence, $\mathcal{A}$ is a σ-algebra.

12 Exercise 2.8

12.1 Solution

Let X be a set and let $\mathcal{A}$ be a σ-algebra on X. Suppose that $\mathcal{A}$ is infinite. Then X must also be infinite, since if X were finite, then $\mathcal{A} \subset \mathcal{P}(X)$ would also be finite.

We will show that $\mathcal{A}$ has uncountably many elements. For this, we first construct a sequence of pairwise disjoint nonempty sets $B_1, B_2, \cdots, \in \mathcal{A}$. Then, we show that $\mathcal{P}(\mathbb{N})$ injects into $\mathcal{A}$.

Step 1:

Claim. For $E \in \mathcal{A}$, if $\{E \cap A : A \in \mathcal{A}\}$ is infinite, then there exists $C \in \mathcal{A}$ such that

$$\emptyset \neq E \cap C \subsetneq E.$$

Proof. Suppose, to the contrary, that no such C exists. Then for every $A \in \mathcal{A}$, we have

$$E \cap A = \emptyset \quad \text{or} \quad E \cap A = E.$$

Hence,

$$\{E \cap A : A \in \mathcal{A}\} = \{\emptyset, E\},$$

which contradicts the assumption that this collection is infinite. ■

Now let $A_1 = X$. Then

$$\{A_1 \cap A : A \in \mathcal{A}\} = \mathcal{A}$$

is infinite.

Suppose that $A_n \in \mathcal{A}$ has been chosen so that

$$\{A_n \cap A : A \in \mathcal{A}\}$$

is infinite. By the claim, there exists $C \in \mathcal{A}$ such that

$$\emptyset \neq A_n \cap C \subsetneq A_n.$$

Then both $A_n \cap C$ and $A_n \cap C^c$ are nonempty members of $\mathcal{A}$.

At least one of the two collections

$$\{(A_n \cap C) \cap A : A \in \mathcal{A}\} \quad \text{and} \quad \{(A_n \cap C^c) \cap A : A \in \mathcal{A}\}$$

must be infinite; otherwise $\{A_n \cap A : A \in \mathcal{A}\}$ would be finite.

Choose A_{n+1} to be one of $A_n \cap C$ and $A_n \cap C^c$ whose associated collection is infinite.

By induction, we obtain a strictly decreasing sequence

$$A_1 \supsetneq A_2 \supsetneq \cdots$$

of nonempty sets in $\mathcal{A}$.

Define

$$B_n = A_n \setminus A_{n+1}.$$

Then each B_n is a nonempty member of $\mathcal{A}$, and the sets $B_1, B_2, \dots$ are pairwise disjoint.

Step 2:

For each $N \subset \mathbb{N}$, define

$$E_N = \bigcup_{n \in N} B_n.$$

Since $\mathcal{A}$ is a σ-algebra, we have $E_N \in \mathcal{A}$.

Moreover, since the sets B_n are pairwise disjoint and nonempty, if $N \neq M$, then

$$E_N \neq E_M.$$

Thus the map

$$N \mapsto E_N$$

is an injection from $\mathcal{P}(\mathbb{N})$ into $\mathcal{A}$.

Since $\mathcal{P}(\mathbb{N})$ is uncountable, $\mathcal{A}$ is uncountable.

13 Exercise 2.9

13.1 Solution

(1) Let $x \in \liminf_i A_i$. Then there exists $j \in \mathbb{N}$ such that

$$x \in \bigcap_{i=j}^{\infty} A_i.$$

Thus $x \in A_i$ for every $i \geq j$, and hence $x \in A_i$ for all but finitely many i.

Conversely, let $x \in A_i$ for all but finitely many i. Then there exists i^* such that $x \in A_i$ for every $i \geq i^*$. Hence,

$$x \in \bigcap_{i=i^*}^{\infty} A_i \subset \liminf_i A_i.$$

Therefore,

$$\liminf_i A_i = \{x : x \in A_i \text{ for all but finitely many } i\}.$$

Let $x \in \limsup_i A_i$. Then for every $j \in \mathbb{N}$, we have

$$x \in \bigcup_{i=j}^{\infty} A_i.$$

But if $x \in A_i$ for only finitely many i, then there exists i^* such that $x \notin A_i$ for every $i \geq i^*$. It follows that

$$x \notin \bigcup_{i=i^*}^{\infty} A_i,$$

which is a contradiction. Therefore, $x \in A_i$ for infinitely many i.

Conversely, let $x \in A_i$ for infinitely many i. Then for evert j, we have

$$x \in \bigcup_{i=j}^{\infty} A_i.$$

Hence, we have

$$x \in \bigcap_{j=1}^{\infty} \bigcup_{i=j}^{\infty} A_i.$$

Therefore,
$$\limsup_i A_i = \{x : x \in A_i \text{ for infinitely many } i\}.$$

(2) Define
$$A_n = \Big(-\frac{1}{n}, 1\Big] \text{ if } n \text{ is odd}, \quad A_n = \Big(-1, \frac{1}{n}\Big] \text{ if } n \text{ is even}.$$
For every n, we have
$$\bigcup_{i=n}^{\infty} A_i = (A_n \cup A_{n+1}) \cup (A_{n+2} \cup A_{n+3}) \cup \cdots = (-1,1] \cup (-1,1] \cup \cdots = (-1,1].$$
Therefore,
$$\limsup_i A_i = \bigcap_{j=1}^{\infty} \bigcup_{i=j}^{\infty} A_i = (-1,1].$$
If n is odd, we have
$$\begin{aligned}\bigcap_{i=n}^{\infty} A_i &= (A_n \cap A_{n+1}) \cap (A_{n+2} \cap A_{n+3}) \cap \cdots \\ &= \Big(-\frac{1}{n}, \frac{1}{n+1}\Big] \cap \Big(-\frac{1}{n+2}, \frac{1}{n+3}\Big] \cap \cdots = \{0\}.\end{aligned}$$
And if n is even, we have
$$\bigcap_{i=n}^{\infty} A_i = \Big(-\frac{1}{n+1}, \frac{1}{n}\Big] \cap \Big(-\frac{1}{n+3}, \frac{1}{n+2}\Big] \cap \cdots = \{0\}.$$
Therefore,
$$\liminf_i A_i = \bigcup_{j=1}^{\infty} \bigcap_{i=j}^{\infty} A_i = \{0\}.$$

(3) For the first case, we have
$$\begin{aligned}\chi_{(\liminf A_i)}(x) = 1 &\iff x \in \liminf_i A_i \iff x \in A_i \text{ for all but finitely many } i \\ &\iff \chi_{A_i}(x) = 1 \text{ for all but finitely many } i \\ &\iff \text{there exists } i^* \ s.t. \ \chi_{A_i}(x) = 1 \text{ for every } i \geq i^* \\ &\iff \liminf_i \chi_{A_i}(x) = 1.\end{aligned}$$
Since the two functions take values only in $\{0,1\}$, the equivalence above implies that they are equal for every x. Therefore,
$$\chi_{(\liminf A_i)}(x) = \liminf_i \chi_{A_i}(x).$$

For the second case, we have

$$\begin{aligned}
\chi_{(\limsup A_i)}(x) = 1 &\iff x \in \limsup_i A_i \iff x \in A_i \text{ for infinitely many } i \\
&\iff \chi_{A_i}(x) = 1 \text{ for infinitely many } i \\
&\iff \sup_{i \geq i^*} \chi_{A_i}(x) = 1 \text{ for every } i^* \\
&\iff \limsup_i \chi_{A_i}(x) = 1.
\end{aligned}$$

Therefore,

$$\chi_{(\limsup A_i)}(x) = \limsup_i \chi_{A_i}(x).$$

Part II

Chapter 3

14 Exercise 3.1

14.1 Solution

Let $\{B_i\}$ be a pairwise disjoint sequence of sets in $\mathcal{A}$ and define

$$A_i = \bigcup_{k=1}^{i} B_k.$$

Then A_i is an increasing sequence of sets in $\mathcal{A}$ and

$$\bigcup_i A_i = \bigcup_i B_i.$$

Thus,

$$\mu\left(\bigcup_i B_i\right) = \mu\left(\bigcup_i A_i\right) = \lim_{i\to\infty} \mu(A_i) = \lim_{i\to\infty} \mu\left(\bigcup_{k=1}^{i} B_k\right).$$

Since μ is finitely additive, we have

$$\lim_{i\to\infty} \mu\left(\bigcup_{k=1}^{i} B_k\right) = \lim_{i\to\infty} \sum_{k=1}^{i} \mu(B_k) = \sum_{i=1}^{\infty} \mu(B_i).$$

Hence μ is countably additive, and therefore a measure.

15 Exercise 3.2

15.1 Solution

Let $\{B_i\}$ be a pairwise disjoint sequence of sets in $\mathcal{A}$ and define

$$A_i = \bigcup_{k=i+1}^{\infty} B_k.$$

Then $\{A_i\}$ is a decreasing sequence of sets in $\mathcal{A}$ and

$$A_i \downarrow \emptyset.$$

Thus, by assumption,

$$\mu(A_i) \to 0.$$

Since μ is finitely additive, for every $n \in \mathbb{N}$, we have

$$\mu\left(\bigcup B_i\right) = \mu\left(\left(\bigcup_{i=1}^{n} B_i\right) \cup A_n\right) = \mu(A_n) + \sum_{i=1}^{n} \mu(B_i).$$

Therefore,

$$\sum_{i=1}^{\infty} \mu(B_i) = \lim_{n\to\infty} \sum_{i=1}^{n} \mu(B_i) = \mu\left(\bigcup B_i\right) - \lim_{n\to\infty} \mu(A_n) = \mu\left(\bigcup B_i\right).$$

Hence μ is countably additive, and therefore a measure.

16 Exercise 3.3

16.1 Solution

Clearly, we have $\mu(\emptyset) = 0$.

Next, let $\{A_i\}$ be a pairwise disjoint sequence of sets in $\mathcal{A}$. We first show that at most one of the sets A_i is uncountable.

Suppose, to the contrary, that A_1 and A_2 are uncountable. Since $A_1 \cap A_2 = \emptyset$, we have

$$A_1 \subset A_2^c.$$

Because A_2 is uncountable and $A_2 \in \mathcal{A}$, its complement A_2^c is countable. Hence A_1 is countable, a contradiction.

If every A_i is countable, then $\bigcup_i A_i$ is countable. Therefore,

$$\mu\left(\bigcup_i A_i\right) = 0, \quad \sum_i \mu(A_i) = 0.$$

Otherwise, exactly one of the A_i is uncountable. Then $\bigcup_i A_i$ is uncountable, and

$$\mu\left(\bigcup_i A_i\right) = 1, \quad \sum_i \mu(A_i) = 1.$$

Hence μ is countably additive, and therefore a measure.

17 Exercise 3.4

17.1 Solution

$$\begin{aligned}
\mu(A) + \mu(B) &= \mu\big((A \cap B) \cup (A \setminus B)\big) + \mu(B) \\
&= \mu(A \cap B) + \mu(A \setminus B) + \mu(B) \\
&= \mu(A \cap B) + \mu\big((A \setminus B) \cup B\big) \\
&= \mu(A \cap B) + \mu(A \cup B).
\end{aligned}$$

18 Exercise 3.5

18.1 Solution

First, we have $\sum_n a_n \mu_n(\emptyset) = 0$, since

$$\mu_1(\emptyset) = \mu_2(\emptyset) = \cdots = 0.$$

Next, let $\{A_i\}$ be a pairwise disjoint sequence of measurable sets. Then

$$\sum_n a_n \mu_n \left(\bigcup_i A_i \right) = \sum_n a_n \left(\sum_i \mu_n(A_i) \right) = \sum_n \sum_i a_n \mu_n(A_i) = \sum_i \sum_n a_n \mu_n(A_i),$$

where the last equality follows from $a_n \mu_n(A_i) \geq 0$.

Hence $\sum_n a_n \mu_n$ is countably additive, and therefore a measure.

19 Exercise 3.6

19.1 Solution

First, we have $\nu(\emptyset) = \mu(\emptyset \cap B) = \mu(\emptyset) = 0$.

Next, let $\{A_i\}$ be a pairwise disjoint sequence of sets in $\mathcal{A}$. Then

$$\nu\left(\bigcup_i A_i\right) = \mu\left(\left(\bigcup_i A_i\right) \cap B\right) = \mu\left(\bigcup_i (A_i \cap B)\right) = \sum_i \mu(A_i \cap B) = \sum_i \nu(A_i).$$

Hence ν is countably additive, and therefore a measure.

20 Exercise 3.7

20.1 Solution

We will show that μ is a measure.

First, we have $\mu(\emptyset) = 0$, since

$$\mu(\emptyset) = \lim_{n\to\infty} \mu_n(\emptyset) = 0.$$

Next, let $\{A_i\}$ be a pairwise disjoint sequence of sets in $\mathcal{A}$. Then

$$\mu\left(\bigcup_i A)\right) = \lim_{n\to\infty} \mu_n\left(\bigcup_i A)\right) = \lim_{n\to\infty} \sum_{i=1}^{\infty} \mu_n(A_i).$$

1. Suppose that $\mu_n(A) \uparrow$ for each $A \in \mathcal{A}$. Let
$$c_{ni} = \mu_n(A_i).$$
Since c_{ni} is a non-negative real number that increases in n to $c_i = \mu(A_i)$ for each i, we have
$$\lim_{n\to\infty} \sum_{i=1}^{\infty} \mu_n(A_i) = \sum_{i=1}^{\infty} \mu(A_i).$$
Hence μ is countably additive, and therefore a measure.
2. Suppose that $\mu_n(A) \downarrow$ for each $A \in \mathcal{A}$ and $\mu_1(X) < \infty$. Let
$$c_{ni} = \mu_n(A_i).$$
Note that c_{ni} is a non-negative real number that decreases in n to $c_i = \mu(A_i)$ for each i. Thus, we have
$$\sup_n |c_{ni}| = \mu_1(A_i).$$
Since μ_1 is a measure, then
$$\sum_{i=1}^{\infty} \sup_n |c_{ni}| = \sum_{i=1}^{\infty} \mu_1(A_i) = \mu_1\left(\bigcup_i A_i\right) \le \mu_1(X) < \infty.$$
Therefore, we have
$$\lim_{n\to\infty} \sum_{i=1}^{\infty} \mu_n(A_i) = \sum_{i=1}^{\infty} \mu(A_i).$$
Hence μ is countably additive, and therefore a measure.

21 Exercise 3.8

21.1 Solution

Let $\mathcal{C} = \{A \cup N : A \in \mathcal{A}, N \in \mathcal{N}\}$. First, we will show that $\mathcal{C}$ is a σ-algebra.

(1) $\emptyset, X \in \mathcal{C}$

Since $\emptyset \in \mathcal{A} \cap \mathcal{N}$, we have

$$\emptyset = \emptyset \cup \emptyset \in \mathcal{C}.$$

Also,

$$X = X \cup \emptyset \in \mathcal{C}.$$

(2) If $E \in \mathcal{C}$, then $E^c \in \mathcal{C}$

Let $E = A \cup N$, where $A \in \mathcal{A}$ and $N \in \mathcal{N}$. Then there exists $F \in \mathcal{A}$ such that

$$N \subset F, \quad \mu(F) = 0.$$

Thus, we have

$$E^c = A^c \cap N^c = A^c \cap \Big(F^c \cup (F \setminus N)\Big) = (A^c \cap F^c) \cup \big(A^c \cap (F \setminus N)\big).$$

Since $A, F \in \mathcal{A}$, we have $A^c \cap F^c \in \mathcal{A}$. Moreover, since

$$A^c \cap (F \setminus N) = (F \setminus N) \setminus A \subset F,$$

we have $A^c \cap (F \setminus N) \in \mathcal{N}$. Hence, $E^c \in \mathcal{C}$.

(3) If $E_1, E_2, ... \in \mathcal{C}$, then $\bigcup_i E_i \in \mathcal{C}$

Let $E_i = A_i \cup N_i$, where $A_i \in \mathcal{A}$ and $N_i \in \mathcal{N}$. Then there exists $F_i \in \mathcal{A}$ such that

$$N_i \subset F_i, \quad \mu(F_i) = 0.$$

Since

$$\bigcup_i N_i \subset \bigcup_i F_i, \quad \mu\left(\bigcup_i F_i\right) \leq \sum_i \mu(F_i) = 0,$$

we have

$$\bigcup_i N_i \in \mathcal{N}.$$

Hence, we have

$$\bigcup_i E_i = \left(\bigcup_i A_i\right) \cup \left(\bigcup_i N_i\right) \in \mathcal{C}.$$

Therefore, $\mathcal{C}$ is a σ-algebra.

Next, we will show that $\mathcal{B} = \mathcal{C}$.

Let $E = A \cup N \in \mathcal{C}$. Then $A \in \mathcal{A}$ and $N \in \mathcal{N}$, which implies that $A, N \in \mathcal{B}$. Thus, we have $A \cup N \in \mathcal{B}$. Therefore,

$$\mathcal{C} \subset \mathcal{B}.$$

On the other hand, let $E \in \mathcal{A} \cup \mathcal{N}$. If $E \in \mathcal{A}$, then

$$E = E \cup \emptyset \in \mathcal{C}.$$

And if $E \in \mathcal{N}$, then

$$E = \emptyset \cup E \in \mathcal{C}.$$

Thus, we have

$$\mathcal{A} \cup \mathcal{N} \subset \mathcal{C}.$$

Since $\mathcal{C}$ is a σ-algebra containing $\mathcal{A} \cup \mathcal{N}$, we have

$$\mathcal{B} = \sigma(\mathcal{A} \cup \mathcal{N}) \subset \mathcal{C}.$$

Hence, $\mathcal{B} = \mathcal{C}$.

Finally, we will show that $\bar{\mu}$ is uniquely defined.

Suppose that

$$B = A_1 \cup N_1 = A_2 \cup N_2,$$

where $A_1, A_2 \in \mathcal{A}$ and $N_1, N_2 \in \mathcal{N}$. If $x \in A_1 \setminus A_2$, then

$$x \in A_1 \subset B = A_2 \cup N_2, \quad x \notin A_2 \quad \Rightarrow \quad x \in N_2.$$

Thus, we have

$$A_1 \setminus A_2 \subset N_2.$$

Similarly, we have

$$A_2 \setminus A_1 \subset N_1.$$

Since $N_2 \in \mathcal{N}$, there is a set $B_2 \in \mathcal{A}$ such that $N_2 \subset B_2$ and $\mu(B_2) = 0$. Moreover, since

$$A_1 \setminus A_2 \subset N_2 \subset B_2,$$

we have

$$\mu(A_1 \setminus A_2) = 0.$$

Similarly,

$$\mu(A_2 \setminus A_1) = 0.$$

Since $A_1 = (A_1 \cap A_2) \cup (A_1 \setminus A_2)$ and this union is disjoint, we have

$$\mu(A_1) = \mu(A_1 \cap A_2) + \mu(A_1 \setminus A_2) = \mu(A_1 \cap A_2).$$

Similarly, we have $\mu(A_2) = \mu(A_1 \cap A_2)$. Hence,

$$\mu(A_1) = \mu(A_2).$$

Hence,

$$\bar{\mu}(B) = \mu(A_1) = \mu(A_2),$$

so $\bar{\mu}$ is well-defined.

22 Exercise 3.9

22.1 Solution

First, we consider the case where m and n are finite.

Let

$$\mathcal{A} = \{A \in \mathcal{B} : m(A) = n(A)\}.$$

We will show that $\mathcal{A}$ is a monotone class.

1. If $A_i \uparrow A$ and each $A_i \in \mathcal{A}$, then

$$m(A) = \lim_{i\to\infty} m(A_i) = \lim_{i\to\infty} n(A_i) = n(A).$$

   Hence, $A \in \mathcal{A}$.

2. Suppose that $A_i \downarrow A$ and each $A_i \in \mathcal{A}$. Since m and n are finite, we have

$$m(A_1) = n(A_1) < \infty.$$

   Thus,

$$m(A) = \lim_{i\to\infty} m(A_i) = \lim_{i\to\infty} n(A_i) = n(A).$$

   Thus, we have $A \in \mathcal{A}$.

Therefore, $\mathcal{A}$ is a monotone class.

Next, we will show that $\mathcal{B} = \mathcal{A}$.

Let $\mathcal{C}$ be the collection of all finite disjoint unions of intervals of the forms

$$(a, b], \qquad (-\infty, b], \qquad (a, \infty),$$

where $a, b \in \mathbb{R}$, together with $\emptyset$ and $\mathbb{R}$. Then $\mathcal{C}$ is an algebra and

$$\sigma(\mathcal{C}) = \mathcal{B}.$$

First, for any $a < b$, we have

$$(a, b + 1/n) \downarrow (a, b].$$

Since each $(a, b+1/n) \in \mathcal{A}$ by assumption and $\mathcal{A}$ is a monotone class, it follows that

$$(a, b] \in \mathcal{A}.$$

Next, for any $b \in \mathbb{R}$,

$$(-n, b] \uparrow (-\infty, b].$$

Since $(-n, b] \in \mathcal{A}$ for all sufficiently large n, the monotone class property gives

$$(-\infty, b] \in \mathcal{A}.$$

Since $(a, n) \in \mathcal{A}$ for all sufficiently large n, the monotone class property gives

$$(a, \infty) \in \mathcal{A}.$$

Therefore, m and n agree on every interval used in the definition of $\mathcal{C}$. By finite additivity, they agree on every finite disjoint union of such intervals. Hence,

$$\mathcal{C} \subset \mathcal{A}.$$

Since $\mathcal{A}$ is a monotone class containing $\mathcal{C}$,

$$\mathcal{M}(\mathcal{C}) \subset \mathcal{A}.$$

By the monotone class theorem,

$$\mathcal{B} = \sigma(\mathcal{C}) = \mathcal{M}(\mathcal{C}),$$

and therefore

$$\mathcal{B} \subset \mathcal{A}.$$

Since $\mathcal{A} \subset \mathcal{B}$ by definition, we conclude that

$$\mathcal{A} = \mathcal{B}.$$

Thus,

$$m(A) = n(A) \qquad \text{for every } A \in \mathcal{B}.$$

Now suppose that at least one of m and n is not finite. For $k \in \mathbb{N}$, define

$$m_k(A) = m(A \cap (-k, k)), \quad n_k(A) = n(A \cap (-k, k)).$$

Thus m_k and n_k are finite, since

$$m((-k, k)) = n((-k, k)) < \infty.$$

For any open interval (a, b), we know that $(a, b) \cap (-k, k)$ is an open interval or the emptyset. Thus,

$$m_k((a, b)) = m((a, b) \cap (-k, k)) = n((a, b) \cap (-k, k)) = n_k(a, b).$$

Hence, from finite case, we can say that $m_k(A) = n_k(A)$ for any $A \in \mathcal{B}$.

For any $A \in \mathcal{B}$, defined

$$A_k = A \cap (-k, k).$$

Then we have

$$A_k \uparrow A.$$

Since

$$m(A_k) = n(A_k)$$

for every k, we obtain

$$m(A) = \lim_{k \to \infty} m(A_k) = \lim_{k \to \infty} n(A_k) = n(A).$$

23 Exercise 3.10

23.1 Solution

The statement is false, even when m and n are finite measures.

Consider

(1) $X = \{1, 2, 3\}$
(2) $\mathcal{A} = \mathcal{P}(X)$
(3) $\mathcal{C} = \{\{1, 2\}, \{2, 3\}\}$.

Then $\sigma(\mathcal{C}) = \mathcal{A}$.

Define m, where

(1) $m(\{\emptyset\}) = 0$, $m(\{1\}) = m(\{3\}) = 1$, $m(\{2\}) = 0$,
(2) $m(\{1, 2\}) = m(\{2, 3\}) = 1$, $m(\{1, 3\}) = 2$,
(3) $m(\{1, 2, 3\}) = 2$.

Define n, where

(1) $n(\{\emptyset\}) = 0$, $n(\{1\}) = n(\{3\}) = 0$, $n(\{2\}) = 1$,
(2) $n(\{1, 2\}) = m(\{2, 3\}) = 1$, $n(\{1, 3\}) = 0$,
(3) $n(\{1, 2, 3\}) = 1$.

Then m, n are finite measures (also σ-finite measures), agree on $\mathcal{C}$, but do not agree on $\sigma(\mathcal{C})$.

Part III

Chapter 4

24 Exercise 4.1

24.1 Solution

Recall that the Lebesgue-Stieltjes measure m^* is defined as

$$m^*(E) = \inf \left\{ \sum_{i=1}^{\infty} l(A_i) : A_i \in \mathcal{C} \text{ for each } i \text{ and } E \subset \bigcup_{i=1}^{\infty} A_i \right\},$$

where $\mathcal{C} = \{(a, b] : a, b \in \mathbb{R} \text{ and } a < b\}$ and $l((a, b]) = \alpha(b) - \alpha(a)$ for an increasing right-continuous function α.

We first show that $l = \mu$ on $\mathcal{C}$. If $b > a \geq 0$, since μ is measure,

$$l((a, b]) = \alpha(b) - \alpha(a) = \mu((0, b]) - \mu((0, a]) = \mu((a, b]).$$

The remaining cases, $b \geq 0 > a$ and $0 > b > a$, are proved similarly.

It is straightforward to verify that α is increasing. To verify that α is right-continuous, let $x_n \downarrow x$. By the identity above,

$$0 \leq \alpha(x_n) - \alpha(x) = \mu((x, x_n]).$$

Since $[x, x_1]$ is compact,

$$\mu((x, x_1]) \leq \mu([x, x_1]) < \infty.$$

Since $(x, x_n] \downarrow \emptyset$, by continuity from above,

$$\mu((x, x_n]) \longrightarrow 0.$$

Hence $\alpha(x_n) \to \alpha(x)$, so α is right-continuous.

We will show that $\mu = m^*$ on the Borel σ-algebra of $\mathbb{R}$.

1. $m^*((a, b]) = \mu((a, b])$

   Since $m^*((a, b]) = l((a, b])$ by Proposition 4.8,

$$m^*((a, b]) = l((a, b]) = \mu((a, b]).$$

2. $m^*(E) = \mu(E)$, where $E \in \mathcal{B}$ and E is bounded.

   For every $\epsilon > 0$, there exists a sequence of sets $\{A_n\}$ in $\mathcal{C}$ such that

$$E \subset \bigcup_{n=1}^{\infty} A_n, \quad \sum_{n=1}^{\infty} \mu(A_n) = \sum_{n=1}^{\infty} l(A_n) \leq m^*(E) + \epsilon.$$

   Since μ is a measure,

$$\mu(E) \leq \mu\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \mu(A_n).$$

   Hence, $\mu(E) \leq m^*(E) + \epsilon$. Since ϵ is arbitrary, we have $\mu(E) \leq m^*(E)$.

   For the other direction, suppose that $E \subset (-N, N]$. Then

$$(-N, N] = E \cup ((-N, N] \setminus E).$$

   Since m^* is a measure on $\mathcal{B}$,

$$m^*((-N, N]) = m^*(E) + m^*((-N, N] \setminus E).$$

   Applying the inequality above to $(-N, N] \setminus E$, we have

$$\mu((-N, N] \setminus E) \leq m^*((-N, N] \setminus E).$$

   Since $\mu((-N, N]) = m^*((-N, N])$ by Step 1,

$$\begin{aligned} m^*(E) = m^*((-N, N]) - m^*((-N, N] \setminus E) &= \mu((-N, N]) - m^*((-N, N] \setminus E) \\ &\leq \mu((-N, N]) - \mu((-N, N] \setminus E) = \mu(E). \end{aligned}$$

   Therefore, $m^*(E) = \mu(E)$.

3. $m^*(E) = \mu(E)$, where $E \in \mathcal{B}$

   Let $I_n = (-n, n)$. Since

$$I_n \cap E \in \mathcal{B}, \quad I_n \cap E \uparrow E, \quad m^*(I_n \cap E) = \mu(I_n \cap E) \leq \mu([-n, n]) < \infty,$$

   we have

$$m^*(E) = \lim_{n \to \infty} m^*(E \cap I_n) = \lim_{n \to \infty} \mu(E \cap I_n) = \mu(E).$$

25 Exercise 4.2

25.1 Solution

By the same argument of the proof of Proposition 4.14 (1), there exist a open set G such that

$$A \subset G, \quad m(G \setminus A) < \epsilon/2.$$

Let's construct F. Since

$$A \cap [-n, n] \uparrow A$$

and $m(A) < \infty$, continuity from below gives

$$m(A \cap [-n, n]) \longrightarrow m(A).$$

Therefore, we may choose N sufficiently large that, with $I = [-N, N]$, we have

$$m(A \setminus I) < \frac{\epsilon}{4}.$$

The set

$$I \setminus A = I \setminus (I \cap A)$$

is Lebesgue measurable and has finite measure. Again by Proposition 4.14(1), there exists an open set G' such that

$$I \setminus A \subset G', \quad m(G' \setminus (I \setminus A)) < \frac{\epsilon}{4}.$$

Let

$$F = I \setminus G'.$$

Since I is closed and G' is open, F is closed. Moreover, since $I \setminus A \subset G'$, we have

$$F = I \setminus G' = I \cap G'^c \subset I \cap (I \setminus A)^c = I \cap (I^c \cup A) = I \cap A \subset A.$$

Now, consider $A \setminus F$. Then we have

$$\begin{aligned} A \setminus F &= (A \setminus I) \cup \Big((A \setminus F) \setminus (A \setminus I) \Big) \\ &= (A \setminus I) \cup \Big((A \cap F^c) \cap (A^c \cup I) \Big) \\ &= (A \setminus I) \cup (A \cap F^c \cap I) \\ &= (A \setminus I) \cup (A \cap (I^c \cup G') \cap I) \\ &= (A \setminus I) \cup (A \cap I \cap G') \end{aligned}$$

Note that two sets on the right are disjoint. Furthermore, since

$$A \cap I \subset A \subset A \cup I^c,$$

we have

$$(A \cap I) \cap G' \subset G' \setminus (I \setminus A) = G' \cap (A \cup I^c).$$

Hence

$$m(A \setminus F) \leq m(A \setminus I) + m(G' \setminus (I \setminus A)) < \frac{\epsilon}{4} + \frac{\epsilon}{4} = \frac{\epsilon}{2}.$$

Thus,

$$m(G \setminus F) = m(G \setminus A) + m(A \setminus F) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Therefore, there exist an open set G and a closed set F such that

$$F \subset A \subset G, \quad m(G \setminus F) < \epsilon.$$

]

26 Exercise 4.3

26.1 Solution

We will show that μ^* is an outer measure.

1. Clearly $\mu^*(\emptyset) = 0$.
2. If $A_1 \subset A_2$, then $\mu^*(A_1) \leq \mu^*(A_2)$.

   Every measurable set containing A_2 also contains A_1. Therefore,

   $$\{B \in \mathcal{A} : A_2 \subset B\} \subset \{B \in \mathcal{A} : A_1 \subset B\}.$$

   Taking the infimum over the larger collection gives

   $$\mu^*(A_1) \leq \mu^*(A_2).$$

3. $\mu^*(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \mu^*(A_i)$ whenever $A_1, A_2, ...$ are subsets of X.

   If $\mu^*(A_i) = \infty$ for some i, then the right-hand side is infinite, and the inequality is immediate. Suppose, therefore, that

   $$\mu^*(A_i) < \infty$$

   for every i.

   Given $\epsilon > 0$, for each i, choose $B_i \in \mathcal{A}$ such that

   $$A_i \subset B_i, \quad \mu(B_i) \leq \mu^*(A_i) + \frac{\epsilon}{2^i}.$$

   Then

   $$\bigcup_{i=1}^{\infty} A_i \subset \bigcup_{i=1}^{\infty} B_i, \quad \bigcup_{i=1}^{\infty} B_i \in \mathcal{A}.$$

   Since μ is a measure,

   $$\mu^* \left(\bigcup_{i=1}^{\infty} A_i \right) \leq \mu \left(\bigcup_{i=1}^{\infty} B_i \right) \leq \sum_{i=1}^{\infty} \mu(B_i) \leq \sum_{i=1}^{\infty} \mu^*(A_i) + \sum_{i=1}^{\infty} \frac{\epsilon}{2^i} = \sum_{i=1}^{\infty} \mu^*(A_i) + \epsilon.$$

   Since ϵ is arbitrary, we have $\mu^*(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \mu^*(A_i)$.

Therefore, μ^* is an outer measure.

Next, we will show that μ^* agrees with the measure μ on $\mathcal{A}$.

Let $A \in \mathcal{A}$. Then $\mu^*(A) \leq \mu(A)$, since $A \subset A$.

For the other direction, for every $B \in \mathcal{A}$ with $A \subset B$, we have

$$\mu(A) \leq \mu(B).$$

Taking the infimum over all such B, we have

$$\mu(A) \leq \inf\{\mu(B) : A \subset B, B \in \mathcal{A}\} = \mu^*(A).$$

Hence, $\mu(A) = \mu^*(A)$.

Finally, we will show that every set in $\mathcal{A}$ is μ^*-measurable.

Let $A \in \mathcal{A}$. Then it is enough to show that

$$\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c) \quad \text{for all} \quad E \subset X.$$

Given E, let $B \in \mathcal{A}$ be any measurable set such that

$$E \subset B.$$

Since $A, B \in \mathcal{A}$,

$$\begin{aligned}
\mu(B) &= \mu(B \cap A) + \mu(B \cap A^c) \\
&= \mu^*(B \cap A) + \mu^*(B \cap A^c) \\
&\geq \mu^*(E \cap A) + \mu^*(E \cap A^c).
\end{aligned}$$

Taking the infimum over all such B, we have

$$\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c).$$

Hence, A is μ^*-measurable.

27 Exercise 4.4

27.1 Solution

$$m(\{x\}) = \lim_{n\to\infty} m\left(\left(x - \frac{1}{n}, x\right]\right) \lim_{n\to\infty}\left(\alpha(x) - \alpha\left(x - \frac{1}{n}\right)\right) = \alpha(x) - \alpha(x-).$$

28 Exercise 4.5

28.1 Solution

1. $m(x+A) = m(A)$

   Consider the collection of half-open intervals $\{I_i\}$ that cover A. Since

   $$x + A \subset \bigcup_{i=1}^{\infty}(x + I_i),$$

   we have

   $$m(x+A) \leq \sum_{i=1}^{\infty} l(x+I_i) = \sum_{i=1}^{\infty} l(I_i).$$

   Taking the infimum over all such collections $\{I_i\}$ that cover A, we obtain

   $$m(x+A) \leq m(A).$$

   Conversely, consider the collection of half-open intervals $\{I_i\}$ that cover $x+A$. Since

   $$A \subset \bigcup_{i=1}^{\infty}(I_i - x),$$

   we have

   $$m(A) \leq \sum_{i=1}^{\infty} l(I_i - x) = \sum_{i=1}^{\infty} l(I_i).$$

   Taking the infimum over all such collections $\{I_i\}$ that cover $x+A$, we obtain

   $$m(x+A) \geq m(A).$$

   Hence, $m(x+A) = m(A)$.

2. $m(cA) = |c|m(A)$

   Suppose that $c > 0$.

   Consider the collection of half-open intervals $\{I_i\}$ that cover A. Since

   $$cA \subset \bigcup_{i=1}^{\infty}(cI_i),$$

we have

$$m(cA) \le \sum_{i=1}^{\infty} l(cI_i) = c \sum_{i=1}^{\infty} l(I_i).$$

Taking the infimum over all such collections $\{I_i\}$ that cover A, we obtain

$$m(cA) \le cm(A).$$

Conversely, consider the collection of half-open intervals $\{I_i\}$ that cover cA. Since

$$A \subset \bigcup_{i=1}^{\infty} \left(\frac{1}{c} I_i \right),$$

we have

$$m(A) \le \sum_{i=1}^{\infty} l\left(\frac{1}{c} I_i \right) = \frac{1}{c} \sum_{i=1}^{\infty} l(I_i).$$

Taking the infimum over all such collections $\{I_i\}$ that cover cA, we obtain

$$cm(A) \le m(cA).$$

Hence, $m(cA) = cm(A)$.

To show the case $c < 0$, we first show that

$$m(-A) = m(A).$$

Let $\{I_i\}_{i=1}^{\infty}$ be a collection of half-open intervals covering A, where

$$I_i = (a_i, b_i].$$

Then

$$-A \subset \bigcup_{i=1}^{\infty} [-b_i, -a_i).$$

These reflected intervals are not of the form $(a, b]$. Given $\epsilon > 0$, define

$$J_i = \left(-b_i - \frac{\epsilon}{2^i}, -a_i \right].$$

Then $(-b_i - \epsilon/2^i, -a_i] \subset J_i$, and hence

$$-A \subset \bigcup_{i=1}^{\infty} J_i.$$

Moreover,

$$l(J_i) = b_i - a_i + \frac{\epsilon}{2^i} = l(I_i) + \frac{\epsilon}{2^i}.$$

Therefore,

$$\begin{aligned} m(-A) &\le \sum_{i=1}^{\infty} l(J_i) \\ &= \sum_{i=1}^{\infty} l(I_i) + \sum_{i=1}^{\infty} \frac{\epsilon}{2^i} \\ &= \sum_{i=1}^{\infty} l(I_i) + \epsilon. \end{aligned}$$

Taking the infimum over all half-open interval covers I_i of A, we obtain

$$m(-A) \le m(A) + \epsilon.$$

Since $\epsilon > 0$ is arbitrary, $m(-A) \le m(A)$.

Applying the same inequality to the set $-A$, we get

$$m(A) = m(-(-A)) \le m(-A).$$

Thus, $m(-A) = m(A)$.

Now let $c < 0$. Since

$$cA = |c|(-A),$$

the positive case gives

$$m(cA) = m(|c|(-A)) = |c|m(-A) = |c|m(A).$$

29 Exercise 4.6

29.1 Solution

1. By Exercise 2.9(1), we can denote that

$$B = \limsup_n A_n = \bigcap_{i=1}^{\infty} \bigcup_{n=i}^{\infty} A_n.$$

Since the collection of Lebesgue measurable sets is closed under countable uninons and intersection, B is Lebesgue measurable.

2. Let

$$B_i = \bigcup_{n=i}^{\infty} A_n.$$

Then $B_i \downarrow B$ and each B_i is Lebesgue measurable. Since $m(B_1) \leq m([0,1]) = 1$ and B is Lebesgue measurable,

$$\lim_{i\to\infty} m(B_i) = m(\lim_{i\to\infty} B_i) = m(B).$$

Since

$$m(B_i) = m\left(\bigcup_{n=i}^{\infty} A_n\right) \geq m(A_i) > \delta$$

for each i, we have

$$m(B) = \lim_{i\to\infty} m(B_i) \geq \delta.$$

3. Since $\sum_{n=1}^{\infty} m(A_n) < \infty$, we have

$$\lim_{i\to\infty} \sum_{n=i}^{\infty} m(A_n) = 0.$$

For each n, we have

$$m(B) \leq m(B_n) = m\left(\bigcup_{n=i}^{\infty} A_n\right) \leq \sum_{n=i}^{\infty} m(A_n).$$

Taking the limit as $i \to \infty$, we obtain

$$m(B) = 0.$$

4. Let

$$A_n = \left[0, \frac{1}{n}\right].$$

Then,

$$\sum_{n=1}^{\infty} m(A_n) = \sum_{n=1}^{\infty} \frac{1}{n} = \infty.$$

However, since

$$B = \bigcap_{i=1}^{\infty} \bigcup_{n=i}^{\infty} \left[0, \frac{1}{n}\right] = \bigcap_{i=1}^{\infty} \left[0, \frac{1}{i}\right] = \{0\},$$

we have

$$m(B) = 0.$$

30 Exercise 4.7

30.1 Solution

Define

$$E = (0, \epsilon) \cup (\mathbb{Q} \cap (\epsilon, 1)).$$

Then $E \subset [0, 1]$, and E is measurable.

Since

$$\overline{(0, \epsilon)} = [0, \epsilon]$$

and $\mathbb{Q} \cap (\epsilon, 1)$ is dense in $(\epsilon, 1)$,

$$\overline{\mathbb{Q} \cap (\epsilon, 1)} = [\epsilon, 1].$$

Therefore,

$$\overline{E} = [0, 1].$$

Moreover,

$$m(\mathbb{Q} \cap (\epsilon, 1)) = 0,$$

so

$$m(E) = m((0, \epsilon)) + m(\mathbb{Q} \cap (\epsilon, 1)) = \epsilon.$$

31 Exercise 4.8

31.1 Solution

Claim. There exists countable subset $A \subset F$ such that $\bar{A} = F$.

Proof. If $F = \emptyset$, then let $A = \emptyset$. Suppose, therefore, that $F \neq \emptyset$.

For each pair $p, q \in \mathbb{Q}$ with $p < q$ and

$$(p, q) \cap F \neq \emptyset,$$

choose one point

$$a_{p,q} \in (p, q) \cap F.$$

Let

$$A = \{a_{p,q} : p, q \in \mathbb{Q},\ p < q,\ (p, q) \cap F \neq \emptyset\}.$$

Since $\mathbb{Q}^2$ is countable, A is also countable. And by construction, $A \subset F$. Moreover, since F is closed,

$$\overline{A} \subset F.$$

To prove the reverse inclusion, let $x \in F$ and let $\epsilon > 0$. By the density of $\mathbb{Q}$ in $\mathbb{R}$, we can choose $p, q \in \mathbb{Q}$ such that

$$x - \epsilon < p < x < q < x + \epsilon.$$

Since $(p, q) \cap F \neq \emptyset$, this intersection is nonempty. Hence there exists. Hence, there exists

$$a_{p,q} \in A$$

. It follows that

$$|x - a_{p,q}| < \epsilon.$$

Thus every neighborhood of x intersects A, so

$$x \in \overline{A}.$$

Therefore, $F \subset \overline{A}$.

Hence, $\overline{A} = F$. ■

We now construct the desired measure.

If $F = \emptyset$, define

$$\nu(B) = 0$$

for every Borel set $B \subset [0,1]$. Then ν is a finite measure whose support is $\emptyset$.

Now suppose that $F \neq \emptyset$. Since A is countable, we may write

$$A = \{a_1, a_2, ...\}$$

if A is infinite, or

$$A = \{a_1, \dots, a_N\}$$

if A is finite.

Assign to each a_i a positive weight w_i such that

$$\sum_i w_i < \infty.$$

For example, if A is infinite, take

$$w_i = \frac{1}{2^i},$$

and if A has N elements, take

$$w_i = \frac{1}{N}.$$

Define, for every Borel set $B \subset [0,1]$,

$$\nu(B) = \sum_{\{i : a_i \in B\}} w_i.$$

Thus ν is a measure, and

$$\nu([0,1]) = \sum_i w_i < \infty.$$

Hence ν is finite. Moreover, since $A \subset F$,

$$\nu(F^c) = 0.$$

It remains to show that F is the smallest closed set with this property. Let $G \subset [0,1]$ be closed and suppose that

$$\nu(G^c) = 0.$$

Since every w_i is positive, no point a_i can belong to G^c. Hence

$$A \subset G.$$

Since G is closed,

$$F = \overline{A} \subset G.$$

Therefore, F is the smallest closed set G such that

$$\nu(G^c) = 0.$$

Thus the support of ν is F.

References